

**ANNALS OF MULTIDISCIPLINARY
RESEARCH, INNOVATION AND TECHNOLOGY (AMRIT)****(A peer-reviewed open access multidisciplinary journal)**www.adtu.in/amrit**RESEARCH ARTICLE****TOURISM: GREEN INVESTMENT****Green Investments as a Vital Tool to Maintain the Sustainability of Tourist Destination:
Gangtok, Sikkim - A Case study**Bidisha Roy¹ and Santosh Kumar Upadhyay^{2*}^{1,2}Department of Tourism & Hospitality, Faculty of Commerce and Management, Assam down town University, Guwahati, Assam- 781026, India*Corresponding author: Santosh Kumar Upadhyay, Email: trysantosh1@gmail.com

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Abstract

Tourism serves a pivotal role in economic growth, but the environment and local communities are frequently threatened by its rapid expansion. The article explains the activities of green investments to maintain tourist destinations, focusing on Gangtok in Sikkim, a hill state of India. Gangtok, considered a continuously developing tourist destination, needs to maintain a balance between its environmental preservation, social progress, and economic growth carefully. A case study approach has been applied and the cost and benefit analysis also compounded in this research article. Secondary data regarding tourism development was generated from different origins. Data about tourism expansion, different indicators of the environment, and economic parameters were analysed to recognize the present scenario and probable threats. The analysis confirms the significance of green investments for the conservation and feasibility of tourism destinations. Eco-friendly infrastructure, community-based tourism initiatives, renewable energy projects, and waste management systems all were considered as green investment projects. Additionally, these initiatives also lowered environmental degradation and improved Gangtok's desirability as a sustainable tourist node. The research explained the fiscal growth generated from green investments which include employability, sound infrastructure, and advance income level. Social capaciousness can be investigated by elevating the local community's participation level and responsible tourism practices. Sustainable tourism enhancement is reflected in the uplifted community engagement, responsible tourism practices, and cultural preservation. The case study disclosed Gangtok's current scenario, probable benefits, and various environmental impact factors for green investment implementation. It also highlights how vital it is for local communities, businesses, and government agencies to create an all-inclusive and operational sustainable framework. The outcomes of the analysis can be used by policymakers to formulate concrete plans to motivate green investments in different tourist locations to ensure a healthy balance between economic growth and environmental sustainability.

Keywords: *Green investment, sustainability, environmental impact, destination management, cost and benefit***1 Introduction**

Tourism is considered as an industry that is very crucial for many places and provides a significant contribution to economic development. Thus, the rapid expansion of tourism many times

becomes a threat for the well-linked tourist nodes. The acceptance of green investments becomes obvious in the context as it serves as a pivotal strategy to generate a poise between the requirement

of environmental conservation and the economic boons of tourism (1). The research area of study is Gangtok, the capital of Sikkim in India, is an excellent example of tourist destination that requires to unify sustainability with the visitor's appeal. Gangtok is situated amongst the Eastern Himalayan range and broadly known for its mesmerizing beauty, varied culture and diverse wildlife (2). To balance the delicate biology of this area, strategy reinforcement is becoming crucial with the rise of tourism. Green investments are being examined as an implicit option due to the crucial significance of protecting Gangtok's varied ecology and the rapid growth of public consciousness of sustainable practices. Financial contributions to programs and projects that advance social responsibility and environmental sustainability are referred to as "green investments" (3). These include a variety of initiatives, such as waste management plans and renewable energy projects, as well as environmentally friendly infrastructure and locally focused tourism. The goal of this case study is to evaluate the factors affecting the environment of Gangtok and to minimize those through green initiatives to support Gangtok's continued viability as a tourism destination. It explores the particular initiatives carried out, the cooperation between local government, companies, and communities, and the effects of these investments on the local economy, environment, and general tourist experience (4).

This study proposes to provide insight into the costs and benefits related to incorporating green investments into the tourism industry by a thorough analysis of legislation, efforts, and results. The results can be a useful guide for other tourism locations facing comparable sustainability issues, providing a model for striking a balance between tourism-driven economic expansion and environmental preservation (5). This research focuses to reinforce practical methods for protecting the most valued tourist locations worldwide by examining the case study of Gangtok and add the larger preservation of sustainable tourism (6). The current research focuses on examining the function of green fiscal policies for improving tourist growth through carbon emission alleviation in China, based on climate importance. Based on the research topic, Data Envelopment analysis was used in the study to derive the study model's efficacy in this context. The research shows that the travellers were motivated to visit climate-supporting stations by China's local tourist nodes, which is popular for its health and wellness tourism. The study's conclusions supported the notion that it is essential to use green finance to alleviate climate change in a famous Chinese tourist node. Empirical results established that by addressing the associated problems, green finance is reducing climate change and boosting tourism growth in Chinese environments. Based on these conclusions, the study produced useful recommendations for Chinese officials, green finance organizations, and climate change policymakers about the growth of the tourism industry (4).

In recent years, environmentally friendly ventures have gained a lot of attention, prompting organizations and companies in sectors like oil and tourism to consider sustainable options. The UNWTO outlines six goals for green investments in the travel sector: strengthening profit margins, satisfying customer preferences, reducing reputational risk, future-proof investment plans, securing sustainable energy sources, and involving local community

members in decision-making processes. Investing in green tourism requires ongoing engagement and work, with the return on investments leaving a lasting impact on the environment, people, and tourists (6).

Sustainable development discourse centres on striking a balance between economic and environmental concerns; responsible tourism (RT) activities play a part in this "greenification" of economies. In India, Kerala is the leading state in the applications of Responsible Tourism. This research investigates various responsible tourism efforts in the different stages of development for the growth of green tourism which is a part of sustainable tourism. The analysis has been driven by the bottom-line approach (TBL) for green economic growth. As per the research, the first phase of responsible tourism was not successful whereas the second and third phases showed greater promise. Further research needs to be carried out in culturally sound remote locations to employ the necessary strategies for the worldwide growth of sustainable tourism (7). Ecological issues brought about by globalization highlight the necessity of environmental management. The development of sustainable tourism, the identification of national park goals, and employee pro-environmental behaviours are all positively impacted by corporate social responsibility (CSR) perception and attitude, according to a study that examined the relationship between CSR perception and attitude and sustainable tourism development in Chinese tourism enterprises. The association between sustainable tourism development and environmental perception and attitude toward CSR is mediated by national park aim identification and pro-environmental staff conduct (8).

The research work was carried out with the objectives to identify the factors affecting the environment of Gangtok in an effort to minimize the factors affecting the environment of Gangtok, by applying green investment techniques and to analyse the future perspective of Gangtok after the implementation of green investment techniques.

2 Research Methodology

2.1 Study Area

Gangtok which is located in southeast Sikkim in the lower Himalayas at an elevation of 1,780 meters (5,480 feet) also the state capital and the administrative centre of the East Sikkim district. The latitude of Gangtok, is 27.33° N 88.62° East, with an average height of 1437 meters (4714 feet). The small town is situated on the slope of a hill, with a palace which is located at a height of 6000 feet (1828 meters). Kanchenjunga is in the East of the city and is also considered as the third tallest snow-covered peak in the world.

2.2 Data Source

Secondary data were collected from different journals, articles, book chapters, and also from different websites. The entire population of Gangtok was considered as the population size of this research. Data has been collected through survey methods and secondary sources.

3 Findings and Analysis

This research was based on COST & BENEFIT analysis to discover the green initiatives taken in Gangtok to preserve the ecology and tourism industry. The tourism industry is completely dependent on the tourists, tourists' behaviour, and their intentions to embrace and maintain green practices throughout the city. An assessment has been done on the cost and benefit analysis to provide the command of green investments to preserve the environmental quality in the future.

3.1 Social Cost

Gangtok is enormously affected by the social cost of tourism (9). Gangtok is a small hill city, but due to heavy tourist falls, the local people started to lose their traditional values, customs, and lifestyles. Additionally, unnecessary crowding and congestion hamper the peace of community people. Sometimes tourists are also cheated and harassed by the local people. Due to the high demand of tourists, local land is also captured by the hotel owners for permanent construction, which is a big threat to the city.

3.2 Environmental Cost

The environment is a big concern for Gangtok's tourism. Vigorous construction of hotels, betterment of the transportation system continuously degrading the environment through air pollution, and noise pollution. Deforestation and the extensive usage of limited water and land lead to a threat to wildlife, and also a big loss of natural beauty. High carbon emissions affecting the air quality and also the uncontrolled footfall of tourists directly contribute to the increase in waste pollution (10).

3.3 Economic Cost

Due to the permanent construction of roads, airports, hotels, shopping malls, and restaurants, inflation (11) is seen in the property value of Gangtok. Local communities feel it is so difficult to cope with these changes. As Gangtok is a tourism-oriented destination, employability remains high in the peak seasons, but in the rest of the times of the year, many people remain unemployed, which hugely affects the city's as well as the state economy.

Along with these costs, benefits also possess in Gangtok like –

- Gangtok attracts lots of international tourists every year and generates foreign currency through them, which boots up the underdeveloped community services and facilities.
- Cultural exchange also takes place through tourists and the preservation of local culture is possible by arrangement of different local festivals seasonally.
- Though environmental degradation is heavily in Gangtok, still govt. is benefitting through these roads, airports, restaurants, and hotels by generating a good amount of income tax. Local people are getting more job opportunities through these businesses.
- To boost the tourism sector of Gangtok, preservation and conservation of natural resources, and wildlife are taken into serious consideration.

4 Suggestions and Recommendations

Gangtok can sustain its tourism and the environment by adopting green initiatives which are essential and highly required for the city's future. All the different costs incurred due to tourism can be replaced by adopting green mechanisms for future sustainability.

Green Initiatives

Gangtok is a small hill station, surrounded by hills and natural beauty, as tourism is the main industry in Gangtok, still to maintain the nature and wildlife of Gangtok, tourist footfalls must be controlled in fragile areas. It will be easier to track the limited number of tourists in fragile areas and the demand for permanent construction like - hotels, resorts, malls, hospitals, banks will also get lower at the same time. Not only that controlled number of tourists will also create less carbon emissions comparatively than the uncontrolled number of tourists. Ropeways, lifts in different places, and malls will create a huge difference in the transport system. Sikkim is the only state without rail connectivity till today, so the railways are essential for Gangtok to control further road constructions and vehicles. Automatically it will help to lower the carbon emissions coming out of vehicles and will enhance the air quality. Liquid fuels like- petrol and diesel can be replaced by hydro fuel which is safer for nature.

An erratic change will be there in Gangtok's pollution level with the adoption of policies like the plastic Ban whether it is good plastic or bad plastic throughout the city. Automatically bio-degradable stuff like jute bags, and carry bags will get prominence and it will lower the soil degradation level. Gangtok's finite water resource can be maintained through rainwater preservation, maximizing the utilization of natural springs and lake water without polluting them due to the delicate biodiversity of the city. The massive tourist gatherings in Gangtok will directly affect the waste pollution and to fix the problem accurate sewage treatment and reuse of wastewater techniques are of utmost importance.

People must be more prone to take organic foods, and fruits instead of nonvegetarian foods, organic foods take less time to cook and at the same time, energy can be saved by utilizing solar cooking systems. By adopting these techniques, the usage of gaseous fuel, biofuel, and electricity can be saved at one time. The entire city's electricity can be replaced by solar power to sustain electric power for future generations. Seasonal flowers, and bushes can be planted and used as road dividers, which not only beautify the entire hill city but also level up the greenery, and enhance the air quality by increasing the oxygen level. Preference must be provided to local delicacies instead of multi-cuisine dishes in different eateries to promote and preserve the traditional culture and customs in front of tourists. These green initiatives can bring a revolutionary transformation in Gangtok's tourism to sustain the environment and natural beauty for future generations.

5 Conclusion

Examining green investments as a means of preserving the sustainability of tourism spots -with a particular emphasis on Gangtok, Sikkim reveals both obstacles and encouraging possibilities. Due to Gangtok's special status as a rapidly developing tourist destination in the Himalayas, environmental preservation and tourism growth must be carefully balanced. The case study emphasizes how crucial it is to implement sustainable practices in order to protect the natural and cultural assets of the area. The article emphasized a number of important points like tourism growth, role of green investments, present sustainable initiatives, etc. Gangtok's constantly growing population has become a threat for its fragile economy and also a severe strain for already finite infrastructure. Responsible growth management is crucial to avoid the negative effects on climate and people. Gangtok has started to execute a balance between environmental-social responsibility and economic growth, as it has already started a number of sustainable tourism projects. To rectify the cons and increase the impact, continuous evaluation of these sustainable developmental programs must be implemented to ensure its effectiveness. There is a lot of scope in Gangtok's tourism industry for application of various green investments. By lessening the carbon footprint of tourists, it also creates a huge effect on the waste management systems, renewable energy projects and biodiversity protection programs. The research focuses on the importance of green investments in Gangtok's tourism industry for sustainability. It implies that financial assistance and partnership among all the stakeholders can generate a sustainable tourist model which can protect the bio diversity and cultural richness through supporting the local economy. Gangtok's sustainability is completely based on adaptive management, continuous monitoring and cooperation among stakeholder's sustainable tourism policies and practices.

Conflict of Interest

The authors declare that there is no conflict of interest in conduction of the work and publication of this paper.

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